



Sheet 2: File Formats and Replicas

1. Consider the following JSON file:

```
{
  "menu": {
    "header": "SVG Viewer",
    "items": [
      { "id": "Open" },
      { "id": "OpenNew", "label": "Open New" },
      null,
      { "id": "ZoomIn", "label": "Zoom In" },
      { "id": "ZoomOut", "label": "Zoom Out" },
      { "id": "OriginalView", "label": "Original View" },
      null,
      { "id": "Quality" },
      { "id": "Pause" },
      { "id": "Mute" },
      null,
      { "id": "Find", "label": "Find..." },
      { "id": "FindAgain", "label": "Find Again" },
      { "id": "Copy" },
      { "id": "CopyAgain", "label": "Copy Again" },
      { "id": "CopySVG", "label": "Copy SVG" },
      { "id": "ViewSVG", "label": "View SVG" },
      { "id": "ViewSource", "label": "View Source" },
      { "id": "SaveAs", "label": "Save As" },
      null,
      { "id": "Help" },
      { "id": "About", "label": "About Adobe CVG Viewer..." }
    ]
  }
}
```

a. Write a program to convert it to

- MessagePack
- Avro
- Proto Buffer
- Thrift

Note: Do not remove nulls and remove white spaces. Append the code at the end of your deliverable.

b. What are the sizes of the new files compared to the original one?



- c. Repeat the same entry in this JSON file 100 times, and rerun your code that converts JSON into the four formats above. What are the new sizes of the new files compared to the new JSON file?

Kindly fill the following table with your results

Format	MessagePack	Avro	Protobuff	Thrift
Original Size				
1 message file size				
100 messages file size				

2. Consider the following avro schema:

```
record Employee {
  string address = "Egypt"
  long salary;
  string name;
}
```

If we changed the schema into each of the following cases, What is the state of Backward compatibility and Forward compatibility, and Why?

- a.

```
record Employee {
  string name;
  string family_name;
  union { null, long }
  salary; long age;
}
```

- b.

```
record Employee {
  long salary;
  string name;
}
```



c.

```
record Employee {  
    string name;  
    boolean active = true;  
    long salary;  
}
```

- d. There are some tools (online web-sites or libraries) that can do the compatibility checks programmatically, please use or install one and mention its name here, then compare your manual solutions with the output of these tools. P.S. if you find a difference between your solution and the output of these tools, do not change yours and discuss the reason with the TA.

Note: Append the code at the end of your deliverable and screenshots for running any external tools.

~~3. Consider a system with Asynchronous Replication. There is a problem when a user is accessing your service from multiple devices, for example a desktop web browser and a mobile app. The user enters a comment using his mobile, then when he opens the desktop web browser there is nothing.~~

- ~~a. Sketch a diagram showing why that happens.~~
- ~~b. Suggest a solution to provide cross device read after write consistency.~~
- ~~c. What are the pros and cons of the solution that you suggested?~~

3. **(Research)** One of the most widely used data file formats in the Hadoop ecosystem is Apache Parquet. Research and answer the following questions:

- a. Is Parquet a columnar-oriented, row-oriented, or hybrid. Explain.
- b. List and briefly explain 3 different data encoding techniques used in Parquet.
- c. Where is Parquet file metadata located in the file?